

Restoring PBR Material Layers from Photoreal Capture Primitives

A screen-space intrinsic decomposition and UV-voting pipeline for Gaussian splats, scans, and extracted meshes.

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Prototype: `pbr_surface.py` + `intrinsic_ab_test.py`

Interactive demo: 127.0.0.1:7862

Abstract. Gaussian splatting and photogrammetry can reproduce a scene with high photographic fidelity, but the captured appearance is usually baked radiance rather than a renderer-native material model. This prototype converts paired screen captures into UV-space PBR material atlases by preserving the relationship between screen-space observations and texture coordinates. After an initial towel smoke test, we reran the pipeline on multi-material Google Scanned Objects assets. The single-view examples exposed the real problem rather than solving it: material labels from one view are sparse and often wrong. The corrected path uses split-source processing plus overlapping multiview capture: use intrinsic decomposition for de-lit texture values, tag materials from original RGB views, vote all labels into UV space, then complete the sparse atlas with a UV-occupancy-constrained resistance fill.



Pipeline diagram

Figure 1. The core design keeps all AI predictions tied to a UV/object-coordinate pass, so screen-space intrinsic estimates and material tags can be written back into texture space.

1. Motivation

Photoreal capture primitives are strong at reproducing the lighting conditions under which they were observed. That strength becomes a weakness when an application needs new lights, edited materials, dynamic time of day, gameplay illumination, or physically plausible asset

reuse. Gaussian splats encode radiance-like appearance in point or ellipsoid attributes; scanned meshes often carry a single baked diffuse texture; extracted surfaces from splats or dense captures may inherit colors that include shadows, highlights, and camera exposure.

Can we take a Gaussian splat, extracted mesh, or scanned textured mesh, screenshot it in world space while preserving texture/UV correspondence, and recover a usable PBR material layer stack?

2. Method

The pipeline keeps every screen-space prediction grounded by an explicit UV pass.

Stage	Input	Output	Purpose
Render/capture	Mesh, splat-derived mesh, or scanned asset	*_color.png , *_uv.png	Capture visible appearance and texture-coordinate relation.
Intrinsic/de-lighting	Color capture	Albedo candidate, optional shading/roughness/metal/specular	Remove lighting from material decisions.
Region segmentation	Source color capture, or albedo for ablations	Connected regions or external masks	Split material-relevant surface regions.
VLM material tagging	RGB region crops	Material labels and texture cues	Convert visual regions into semantic PBR priors before color cues are washed out.
UV vote	De-lit albedo, RGB labels, UV pass	UV-space albedo atlas and material map	Preserve renderer-native

Stage	Input	Output	Purpose
			texture coordinates.
UV completion	Sparse albedo, material map, UV occupancy mask	Completed atlas domain	Fill voids while resisting texture edges and empty atlas padding.
PBR synthesis	Albedo + material labels	Normal, height, ORM	Generate a renderer-consumable material set.

Capture pair and UV projection

Figure 2. A color capture and an aligned UV pass let image-space predictions be projected back into the texture atlas.

3. Backend Selection Smoke Test

The initial validation object was a towel asset from Google Scanned Objects. Five intrinsic/de-lighting backends were run through the same harness, and successful outputs were baked to UV space. This chose Marigold Appearance as the practical default, but the towel was not a strong material-divergence test.

Backend	Produced channels	Runtime	Delta vs. RGB	Observation
Identity baseline	Albedo copy, grayscale shading	0.1 s	0.000	No de-lighting; useful control.
Marigold Lighting	Albedo, shading, residual	725.3 s	0.043	Strong lighting separation, but washed out woven detail.

Backend	Produced channels	Runtime	Delta vs. RGB	Observation
Marigold Appearance	Albedo, roughness, metallicity	423.4 s	0.159	Best default in this test; preserved textile regions.
IntrinsicAnything	Albedo, specular	56.0 s	0.548	Fast after setup; lifts background and compresses contrast.
Intrinsic Image Diffusion	Albedo, roughness, metal	564.6 s	0.297	Successful but gray/brown relighting bias.

A/B comparison grid

Figure 3. End-to-end comparison of intrinsic backends, UV-baked albedo, material map, and auxiliary outputs.

Backend metrics chart

Figure 4. Runtime and albedo-change summary. Delta is a smoke-test proxy, not a physical accuracy score.

4. Multi-Material Validation

The towel smoke test was replaced by scanned objects with clear material divergence: red scissors, a cushion-grip screwdriver, and a snap-lock can opener. The single-view versions are useful diagnostics, but not a complete solution. They show why one view is insufficient: the UV atlas remains sparsely sampled, screen-space occlusion hides important materials, and a dominant visible part can overwhelm the material vote. The corrected run uses Marigold Appearance for de-lit albedo values, but performs segmentation and VLM material tagging on original RGB captures.

Split-source material tagging comparison

Figure 5. Tagging from the de-lit albedo can wash away semantic color cues. The corrected run tags the source RGB regions and projects those labels into UV space.

Asset	Expected material contrast	RGB-derived VLM segment labels	Vision calls	Fallbacks	Direct atlas coverage
Red scissors	Plastic handles, steel blades	plastic 12, metal_steel 8	21	0	0.83%
Cushion-grip screwdriver	Plastic grip, steel shaft	plastic 20, metal_steel 12	33	0	0.47%
Snap-lock can opener	Black plastic/rubber, steel cutter	plastic 7, metal_steel 12	20	0	0.91%

Multi-material validation examples

Figure 6. End-to-end multi-material examples. RGB-derived material labels are voted into UV space while Marigold supplies de-lit texture values for PBR synthesis.

The toaster candidate was also tested. It exposed a useful remaining problem: a large reflective unassigned body region was filled as `stone` by the fallback heuristic even though smaller segmented regions were correctly tagged as `metal_steel`. This shows that manufactured-object priors and better region proposal masks matter for appliances and broad glossy surfaces.

4.1 Overlapping multiview voting

We added a generated `sphere32` Blender capture mode that renders 32 overlapping RGB/UV pairs from a Fibonacci sphere around the asset. The existing baker already accumulates material votes per UV texel across all capture pairs, so the multiview fix is a capture protocol change rather than a new atlas algorithm.

On the can opener, the single-view material atlas was front-view biased: 9,158 tagged atlas pixels voted `plastic` and only 416 voted `metal_steel`. The 32-view pass used 384 region-level VLM calls with zero fallbacks and changed the UV vote to 14,973 `metal_steel`, 6,008 `plastic`, and 287 `unknown` pixels. Direct atlas coverage rose from 0.91% to 2.71% at 512px captures. This still is not full texture coverage, but it is a much better validation of the UV voting mechanism.

32-view UV material voting

Figure 7. Single-view material labels are sparse and front-view biased. A 32-view capture set contributes repeated material observations to the same UV atlas and produces a more defensible vote.

Capture protocol	Capture pairs	VLM region calls	Fallbacks	Direct atlas coverage	Tagged atlas coverage	Main issue
Single front view	1	20	0	0.91%	0.91%	Front-view material bias.
sphere32 multiview	32	384	0	2.71%	2.03%	Costly; still sparse at 512px captures.

4.2 UV-space resistance fill

The multiview vote gives higher-confidence samples, but it is still sparse because many atlas texels are never hit by a screen pixel. A naive dilation can fill those holes, but it also spreads across empty atlas padding and material boundaries. The implemented fix is a resistance-weighted UV flood constrained by a target mask. The target mask represents known UV island territory; for this scanned can opener we derived it from non-empty source texture texels, and for splat-derived meshes the same mask should come from a UV occupancy rasterization of the extracted triangles.

For each missing texel, neighboring samples vote with a weight proportional to $\exp(-\text{resistance} * \text{color_difference})$ in a provisional guide image. This allows fills to continue through smooth texture regions while resisting strong albedo or material-edge changes. Material IDs are expanded with the same guide, so labels spread through compatible UV neighborhoods instead of simple nearest-neighbor padding.

On the can opener, direct 32-view coverage was 2.71% of the atlas and direct tagged coverage was 2.03%. The inferred UV occupancy target covered 39.34% of the atlas. A 96px resistance fill covered 35.89% of the atlas, or 91.23% of the UV target, and expanded tagged material pixels to 34.84%. The filled material map contained 266,033 `metal_steel`, 98,414 `plastic`, and 883 `unknown` pixels.

UV-space resistance fill comparison

Figure 8. Sparse 32-view UV votes are expanded inside an inferred UV occupancy mask. The resistance fill completes most of the atlas island domain without treating empty atlas padding as usable material.

UV completion stage	Atlas coverage	Tagged atlas coverage	Notes
Direct sphere32 samples	2.71%	2.03%	Trusted but too sparse for renderer export.
UV occupancy target	39.34%	n/a	Estimated island domain from the scan texture.
96px resistance fill	35.89%	34.84%	Covers 91.23% of the target without filling empty atlas padding.

4.3 PBR atlas generation

Each corrected and completed material map was combined with a de-lit Marigold albedo atlas and converted into renderer-style maps.

Generated map	Role
albedo_baked.png	De-lit UV-space base color candidate.
source_baked_albedo_materials.png	Integer material-label atlas from original RGB regions.
pbr/albedo_baked_n.png	Normal map synthesized with material-aware relief priors.
pbr/albedo_baked_h.png	Height map for parallax/displacement-style workflows.
pbr/albedo_baked_orm.png	Occlusion, roughness, metallicity map.

5. Reproducibility

The interactive comparison app is available locally:

```
python C:\Users\gregor\Documents\CubeBauhausOnline\tools\intrinsic_ab_test.py
```

The current best default is to run Marigold Appearance first, then run material segmentation/tagging from the original RGB capture and combine both products in the atlas stage:

```
python C:\Users\gregor\Documents\CubeBauhausOnline\tools\intrinsic_ab_test.py  
--backends marigold_appearance  
--size 1024  
--device cuda  
--output C:\Users\gregor\Documents\CubeBauhausOnline\artifacts\intrinsic-ab-t
```

Recommended default

```
32x overlapping Color + UV captures -> Marigold Appearance albedo ->  
RGB-connected region segmentation -> Qwen VLM tags -> UV vote -> UV  
occupancy resistance fill -> PBR atlas
```

6. Limitations

This is a prototype smoke test, not a benchmark. Single-view examples are not sufficient for material correctness; overlapping multiview capture is required. Even 32 views at 512px produced only 2.71% direct atlas coverage on the can opener, so a UV-space completion stage is required before renderer export. Roughness, metallicity, height, and normal maps are plausible renderer inputs rather than measured ground truth; VLM material labels can fail on ambiguous or mixed-material crops; and Gaussian splat assets still need a robust way to render or derive coordinate/UV passes and UV occupancy masks.

7. Conclusion

The prototype demonstrates an end-to-end path from photoreal capture appearance to renderer-native PBR maps. The important engineering move is to preserve UV correspondence while applying screen-space AI models. With paired RGB/UV captures, diffusion-based intrinsic decomposition can reduce lighting leakage, VLM tagging can recover semantic material

classes, multiview voting can place those observations into atlas space, and UV occupancy resistance fill can turn sparse trusted samples into conventional albedo, normal, height, ORM, and material-label atlases.

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